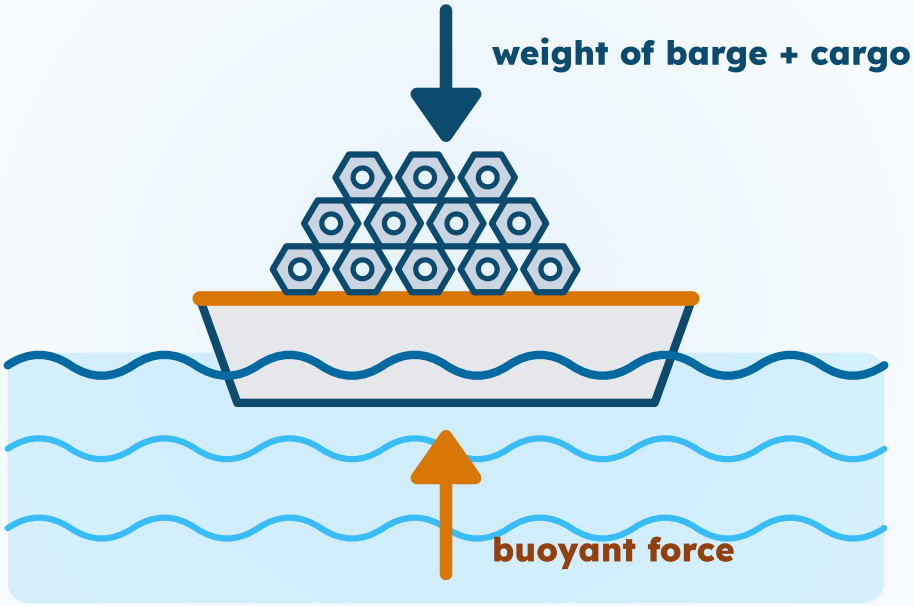
SHIP SHAPE

Engineer's Logbook

Fold it. Seal it. Load it. Float it.



ENGINEER ID

Name: _____

Partner: _____

Event: Ship Shape **Our best cargo:** _____

If found, please return to the DDMES Science Olympiad binder shelf.

How a barge engineer works

Engineers don't guess and hope. They **plan**, **test**, write down what happened, and change **one thing at a time**. Water pushes up on anything in it. A barge floats as long as it can push aside water that weighs as much as the barge and its cargo, so shape matters as much as material.

1
SKETCH
Draw your barge on graph paper. Plan where the tape goes.

2
PREDICT
Do the capacity math. Write your estimate BEFORE testing.

3
BUILD
Kit only, 15 minutes, fits the box.

4
TEST
Goggles on. Load one nut at a time, spread evenly.

5
IMPROVE
Log it, change one thing, graph your best.

The Engineer's Code

- [ ] Goggles on (Category C indirect-vent) before anyone touches the water.
- [ ] Get an adult to say 'Clear!' before a test. Wipe up spills right away.
- [ ] Fold corners. Don't cut them: cuts leak.
- [ ] Write your estimate first, test second.
- [ ] Write every trial down, even the ones that sink fast.
- [ ] Change one thing at a time so you know what worked.

My build kit

- [ ] This logbook and pencils
- [ ] Goggles, scissors, a ruler
- [ ] A calculator (optional)
- [ ] Foil, a plastic sheet, masking tape, straws
- [ ] A tub or sink of water and a towel
- [ ] Pennies or nuts for cargo

Logbook map

Know your event	p. 3-4
Design sketches (graph paper)	p. 5-6
Trial Logs	p. 7-10
Best-cargo graph	p. 11
What went wrong?	p. 12
At-home build challenges	p. 13-16
Tournament checklist	p. 17
Missions and badges	p. 18-19

Where to practice

Kitchen sink
your home test tank

Bathtub
float a plastic bowl and watch it sink lower as you add cargo

Recycling bin
clear plastic sheets, trays, and cartons to try

A lake or harbor
look at how barges and boats sit low when loaded

Grocery store
a 1 liter bottle of water has a mass of about 1 kilogram

Open Lab Fridays
the Ship Shape bin has a tub, clay, and a measuring cup

Did you know? The biggest cargo ships are made of steel and carry more than 20,000 shipping containers. They float for the same reason your foil barge does: a big hull pushes aside a LOT of water.

Know your event: the rules

Pairs get 15 minutes to build a barge from a kit (foil, plastic sheet, 50 cm of tape, 5 straws) that fits in a 15 x 15 x 2.5 cm box, estimate its cargo to 0.01 g, load hex nuts until it sinks, and take a written test on buoyancy (bring goggles, scissors, a ruler, pencils; calculator optional).

Part	What the rules say
Kit	1 foil sheet (12 x 12 in), 1 plastic sheet (10.75 x 10 in), 50 cm of masking tape, 5 straws.
Build time	15 minutes at the tournament.
Size rule	The barge must fit inside a box 15 cm x 15 cm x 2.5 cm tall.
Cargo	Metal hex nuts, added one at a time until the barge sinks (or holds 500 grams).
Test	A written test on floating and sinking.
Safety	Wear Category C indirect-vent goggles while testing.
Barge rules	One cargo area, open at the top for loading. Kit materials only (not even a paper clip from home).
Estimate	Before testing, write how much cargo your barge will hold, to 0.01 g (like 250.00 g).
Written test	40 questions worth half a point each, so the test is out of 20 points. Keep working on it until you are called to test.

How it's scored

Final score	Test points + grams of cargo held + 5 bonus if the estimate is exact.
Tier 1	Barge follows all rules: open top, fits in the box, kit materials only.
Tier 2	Broke a building rule. Still tests, but ranks below every Tier 1 team.
Sunk	The top edge of the cargo area is all the way under water.
Top limit	Loading stops at 500 g.

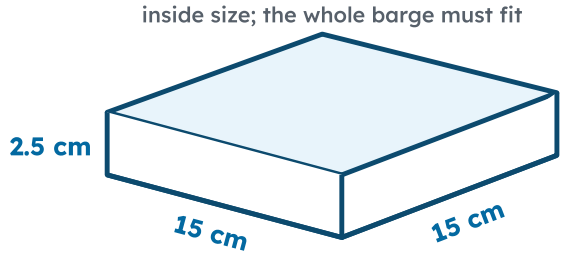
Score math

16 test points
+ 204.50 g held
+ 5 bonus (the estimate was exactly 204.50 g)
= 225.50

Not exact? No bonus, but a close estimate can win a tie.

Tricky rules: if water leaks in but the rim is still above water, the judge keeps loading, and you may **not** scoop water out. A lid over the cargo area (even one with a hole) makes it Tier 2.

The measuring box



Bring to the tournament

- [ ] Category C indirect-vent goggles (regular safety glasses are NOT allowed)
- [ ] Scissors and a ruler
- [ ] Pencils
- [ ] Optional: a basic calculator, a protractor, a tape measure

Know your science: why barges float

Words and big rules

Buoyancy (buoyant force)	The upward push of water (or any liquid) on an object in it.
Displacement	Water being pushed aside by an object. The water level rises.
Float	Stay on top. The upward push is as big as the object's weight.
Sink	Go down. The object's weight is bigger than the upward push.
mL (milliliter)	A unit for measuring water. 1 mL of water has a mass of 1 gram.
Archimedes' principle	The upward push on an object equals the weight of the water it pushes aside.
The gram rule	1 cubic cm of water = 1 mL = 1 gram.
Volume	Length x width x height. Measure in cm to get cubic cm.
Capacity	Volume of the hull below the rim (in g) minus the mass of the barge = most cargo it can hold.
Real-world percent	Actual grams held / predicted grams x 100. Real barges leak, tip, and bend, so they hold less than the math.

Flotation facts

Flotation	The ability to float. An object floats when the buoyant force equals its weight before it is all the way under.
Density	How much mass is packed into a space. Heavy for its size = dense.
Denser than water	Sinks (steel, rocks, clay balls).
Less dense than water	Floats (wood, oil, a boat full of air).
Why ships float	A ship's hull holds a lot of air, so the ship plus the air is less dense than water.

Load it like this

2	8	5
6	1	7
4	9	3

Middle first, then corners and edges evenly, so the barge stays level.

Capacity math (Lesson 3 example)

Cargo area 12 cm x 10 cm x 2 cm = **240 cubic cm** = 240 g of water pushed aside. Barge mass 10 g. 240 - 10 = **230 g** of cargo (in a perfect world).
Real barges leak, tip, and bend. If your barges hold about 80 percent of the math: 230 x 0.8 = **184 g**. If 1 nut is 3.08 g: 184 / 3.08 = 59.7, so about 60 nuts. 60 x 3.08 = **184.80 g**. That's your estimate.

Check yourself

1. Fill in: 1 cubic cm of water = ___ mL = ___ g.

2. A hull is 10 cm x 10 cm x 2.5 cm. What is its volume? How many grams of water can it push aside?

3. A steel nut sinks. Is steel denser or less dense than water?

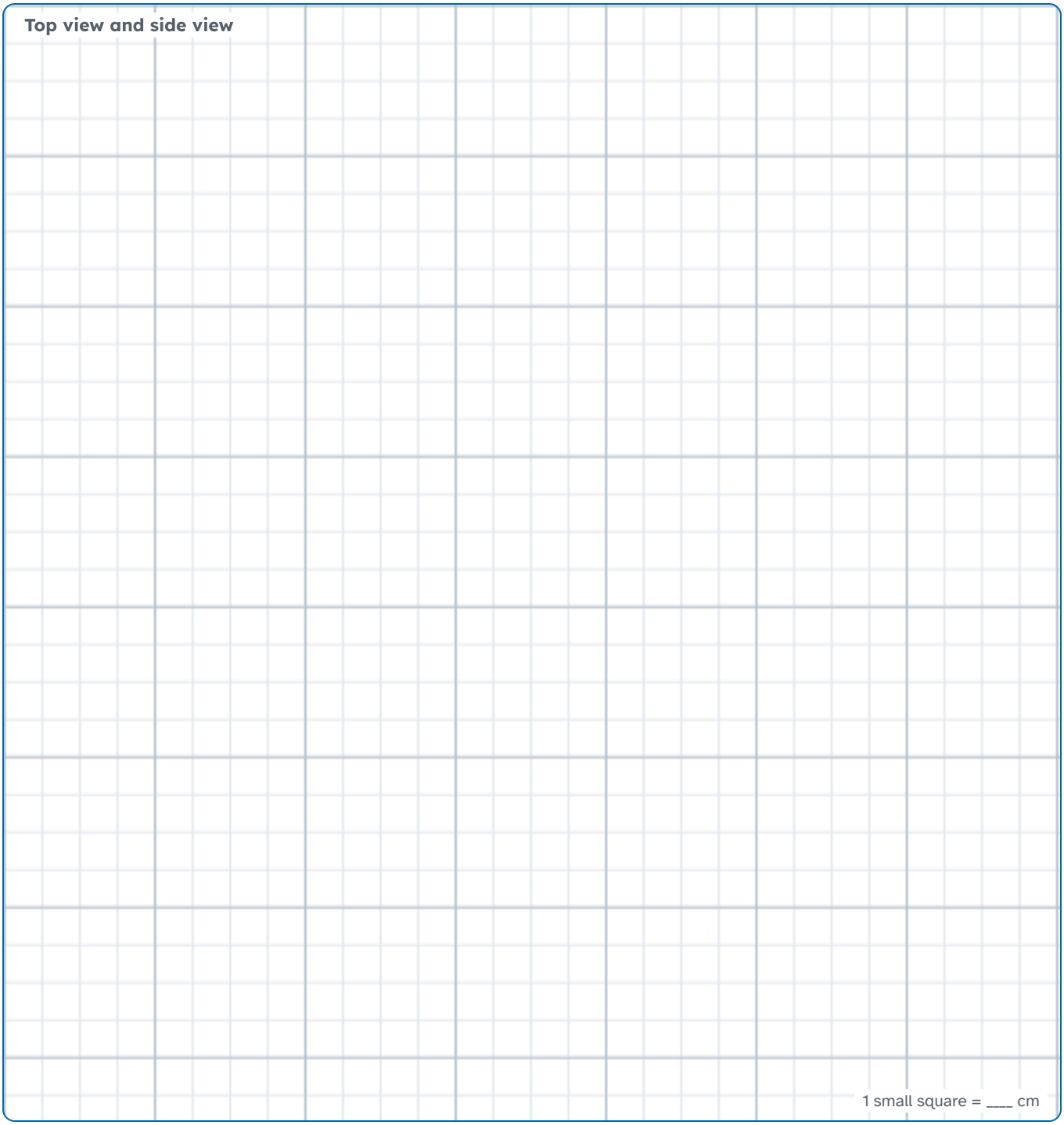
4. Why does a huge steel ship float?

Answers: 1. 1 mL and 1 g. 2. 10 x 10 x 2.5 = 250 cubic cm, so 250 g of water. 3. Denser (it sinks). 4. The hull holds a lot of air, so the ship plus the air is less dense than water.

Design A: Plastic sheet tray, lined or wrapped in foil

Draw your barge from above and from the side. Label the length, width, and wall height in cm, every fold, and where each piece of tape goes (you only get 50 cm). Then do the capacity math below.

Design name _____ **Date** _____ **Version #** _____



Size in cm: L ___ x W ___ x H ___ = _____ cubic cm

Predicted cargo (g) and why this design will hold it:

Design B: Foil hull with straws taped on the rim

Draw your barge from above and from the side. Label the length, width, and wall height in cm, every fold, and where each piece of tape goes (you only get 50 cm). Then do the capacity math below.

Design name _____ **Date** _____ **Version #** _____

Top view and side view

1 small square = ___ cm

Size in cm: L ___ x W ___ x H ___ = _____ cubic cm

Predicted cargo (g) and why this design will hold it:

Trial Log #1

Every barge test is a trial. Write your estimate BEFORE you load. Result: nuts (or pennies) held and the mass in grams.

Date _____ **Design or setup** _____ **Teammates** _____

Trial	What I changed (change 1 thing!)	My estimate (g, to 0.01)	Result (# held, grams)	What I learned
1				
2				
3				
4				
5				
6				
7				
8				

Best trial on this page: # _____ **Why it worked:** _____

Trial Log #2

Every barge test is a trial. Write your estimate BEFORE you load. Result: nuts (or pennies) held and the mass in grams.

Date _____ **Design or setup** _____ **Teammates** _____

Trial	What I changed (change 1 thing!)	My estimate (g, to 0.01)	Result (# held, grams)	What I learned
9				
10				
11				
12				
13				
14				
15				
16				

Best trial on this page: # _____ **Why it worked:** _____

Trial Log #3

Every barge test is a trial. Write your estimate BEFORE you load. Result: nuts (or pennies) held and the mass in grams.

Date _____ **Design or setup** _____ **Teammates** _____

Trial	What I changed (change 1 thing!)	My estimate (g, to 0.01)	Result (# held, grams)	What I learned
17				
18				
19				
20				
21				
22				
23				
24				

Best trial on this page: # _____ **Why it worked:** _____

Trial Log #4

Every barge test is a trial. Write your estimate BEFORE you load. Result: nuts (or pennies) held and the mass in grams.

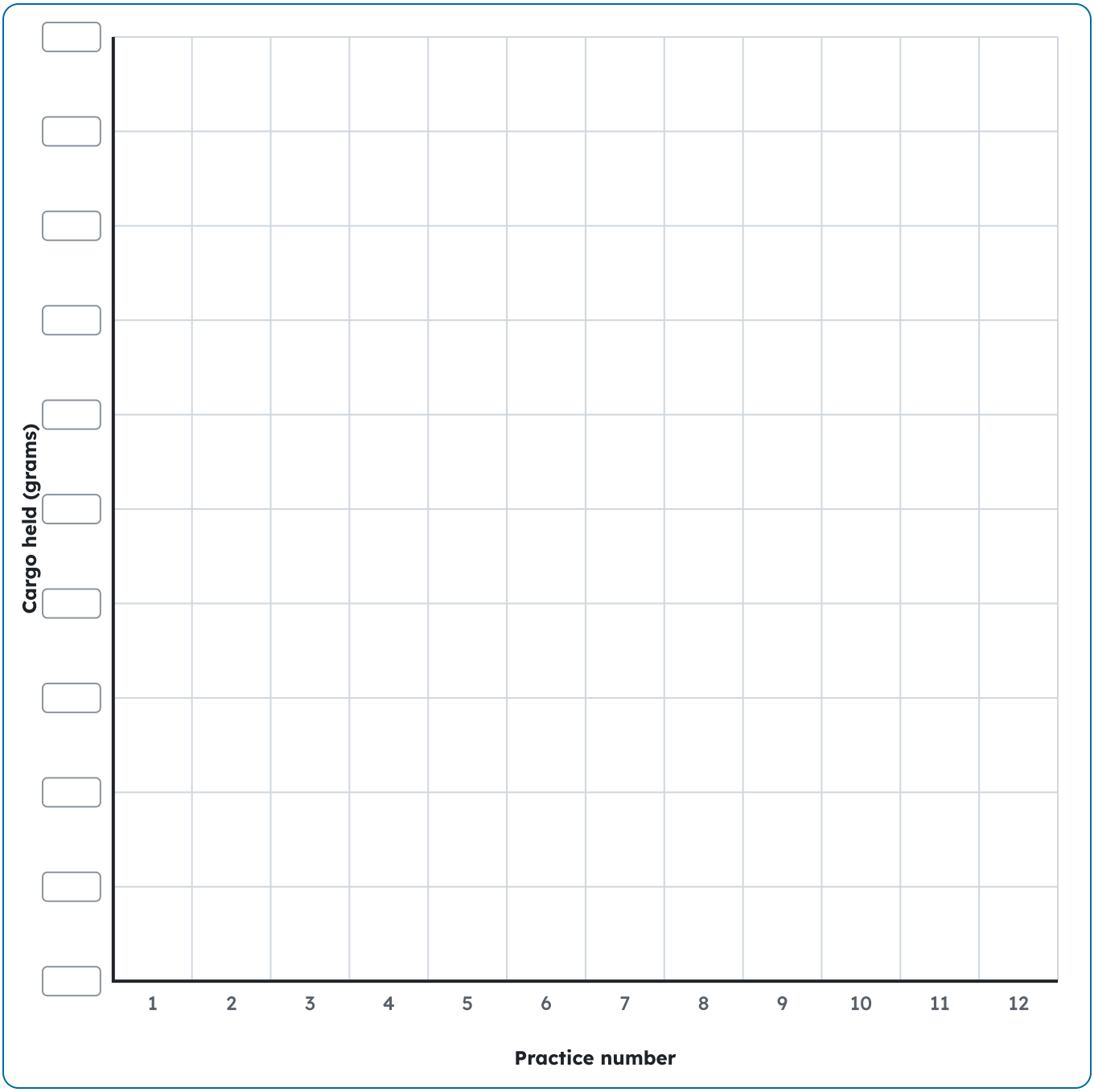
Date _____ **Design or setup** _____ **Teammates** _____

Trial	What I changed (change 1 thing!)	My estimate (g, to 0.01)	Result (# held, grams)	What I learned
25				
26				
27				
28				
29				
30				
31				
32				

Best trial on this page: # _____ **Why it worked:** _____

My best-cargo graph

After each practice day, put a dot for the most cargo your barge held (in grams). Write the numbers on the side (count by 25s or 50s), then connect the dots.



Date of #	1	2	3	4	5	6	7	8	9	10	11	12

Higher is better! Draw a line across at your estimate each time. A dot right on the line means an exact estimate, worth 5 bonus points. The most the judge will load is 500 g.

My best so far: _____

My tournament goal: _____

Is my line going the right way? _____

What went wrong?

Every engineer's design fails sometimes. Good engineers figure out **why**. Here are problems Ship Shape pairs hit a lot, and fixes to try.

Problem	What it looks like	Try this (one fix at a time)
Leaks at the corners	Water seeps in where the walls meet.	Fold corners, don't cut them. Seal seams with tape on the outside.
Ran out of tape	All 50 cm used in one spot.	Plan the tape on your sketch: seams first, then the rim.
Tipped over	One side dips and water pours in.	Load the middle first, then corners and edges evenly.
Bottom sags	The floor bends down in the middle.	Spread the nuts. Stiffen the floor or use the plastic sheet as a tray.
Straws filled up	Straws on the rim fill with water.	Seal straw ends with tape if you use them as floats.
Doesn't fit the box	Too long, too wide, or taller than 2.5 cm.	Measure while you build. Check with the 15 x 15 x 2.5 cm box. Too big = Tier 2.
Estimate way off	Held much more or less than you wrote.	Use your real-world percent from past trials, and write the estimate first.

My fix-it log

Write down each problem you hit. Fix 3 problems to earn the **Leak Stopper** badge.

What went wrong	Why I think it happened	What I tried	Did it work?
			☐ yes ☐ no
			☐ yes ☐ no
			☐ yes ☐ no
			☐ yes ☐ no
			☐ yes ☐ no
			☐ yes ☐ no
			☐ yes ☐ no

AT-HOME BUILD CHALLENGE 1 · AFTER LESSON 1 (FRI OCT 23)

Penny Barge

Penny Loader badge

Your mission: shape a foil boat that fits on a 15 x 15 cm square and see how many pennies it holds before it sinks. Then change the shape and beat your record.

You need

- [ ] 2 sheets of foil, about 12 x 12 in
- [ ] A ruler and paper (draw a 15 x 15 cm square)
- [ ] Pennies (or other coins)
- [ ] A sink, tub, or big bowl of water
- [ ] A towel

Build it

1. Draw a 15 x 15 cm square on paper. Your boat must fit on it.
2. Fold one foil sheet into a boat. Fold, don't crumple. No tape.
3. Float it. Add pennies one at a time, middle first, then spread out.
4. Stop when water pours in. Count the pennies.
5. Make boat 2 with ONE change: taller sides, a flatter bottom, or stronger corners.

Draw both boats: top view and side view

My results

Boat	Shape (describe it)	Pennies held	Grams (pennies x 2.5)
1			
2			
3 (bonus)			

Which boat held more? What was different about its shape?

Safety: wipe up spills so nobody slips. Keep pennies away from little brothers and sisters.

Date done: _____ **Coach or Captain initials (Penny Loader badge):** _____

AT-HOME BUILD CHALLENGE 2 · AFTER LESSON 2 (FRI DEC 4)

Capacity Predictor

Archimedes Ace badge

Your mission: use Archimedes' rule (1 cubic cm of water = 1 g) to **predict** how many pennies a foil tray will hold. Then test it.

You need

- [ ] Foil, about 12 x 12 in
- [ ] A ruler
- [ ] Pennies (dated 1983 or later are 2.5 g each)
- [ ] A sink or tub of water
- [ ] A calculator (optional)

Build it

1. Fold a foil tray with straight sides. Try 10 cm x 10 cm x 2 cm.
2. Measure the inside length, width, and wall height in cm.
3. Volume = L x W x H cubic cm. That's how many grams of water it can push aside.
4. Predicted pennies = volume / 2.5. Write it down BEFORE testing.
5. Load pennies evenly until it sinks. Real-world percent = actual / predicted x 100.

Sketch your tray and label its measurements

My results

L x W x H (cm)	Volume (cubic cm)	Predicted pennies	Actual pennies	Real-world %

Did your tray hold less than the math said? What do you think got in the way (leaks, tipping, bending)?

Safety: wipe up spills right away. Water and floors don't mix.

Date done: _____ **Coach or Captain initials (Archimedes Ace badge):** _____

AT-HOME BUILD CHALLENGE 3 · AFTER LESSON 3 (FRI JAN 22)

15-Minute Home Kit Build



15-Minute Builder badge

Your mission: make a home kit like the tournament kit, build a barge in **15 minutes**, check that it fits the box, and test it.

You need

- [ ] Foil, 12 x 12 in
- [ ] A clear plastic sheet (a sheet protector or a cut report cover)
- [ ] 50 cm of masking tape (measure it!)
- [ ] 5 straws
- [ ] Scissors, a ruler, a timer, pennies, and a tub of water

Build it

1. Make a check box: a cardboard box 15 x 15 cm and 2.5 cm tall (or a square and a ruler).
2. Set a 15-minute timer. Build your barge from the kit only.
3. Box check: does it fit all the way inside? If not, it would be Tier 2.
4. Write your estimate. Then load pennies evenly until the rim goes under.
5. Write your build steps as a numbered list on the grid so you can do it faster.

My numbered build steps (or a sketch)

My results

Build time (min)	Fits the box? (Y/N)	Estimate (g)	Pennies held	Grams held

What slowed you down? What will you do first next time?

Safety: an adult helps with scissors. Goggles on for the test if you have them.

Date done: _____ **Coach or Captain initials (15-Minute Builder badge):** _____

AT-HOME BUILD CHALLENGE 4 · AFTER LESSON 4 (FRI FEB 19)

Estimate to the 0.01 g



Estimate Expert badge

Your mission: be the judge at home. Write an estimate to 0.01 g, load your barge, and score yourself like the tournament.

You need

- [ ] A barge from your home kit (or a new one)
- [ ] Pennies (2.50 g each)
- [ ] A tub of water and a towel
- [ ] Your capacity math and real-world percent
- [ ] A family member to be the judge

Build it

1. Do the capacity math. Multiply by your real-world percent.
2. Round to a whole number of pennies. Estimate = pennies x 2.50 g. Write it like 180.00 g.
3. The judge loads pennies one at a time, middle first, then evenly.
4. Stop when the rim goes all the way under. Grams held = pennies x 2.50.
5. Score = grams held + 5 if the estimate was exact.

Show your estimate math here

My results

Try	Estimate (to 0.01 g)	Grams held	Difference (g)	Bonus (5 if exact)
1				
2				

How close was your estimate? What will you change in your math before the tournament?

Safety: wipe up spills right away. Put your goggles in your tournament bag when you are done.

Date done: _____ **Coach or Captain initials (Estimate Expert badge):** _____

Tournament-day checklist

Sat Mar 13, Chatham Regional (Seaforth High School). Check each box as you go.

Pack your bag

- [ ] Category C indirect-vent goggles (regular safety glasses are NOT allowed)
- [ ] Scissors and a ruler
- [ ] Pencils
- [ ] A calculator (optional)
- [ ] A water bottle and a snack

The night before

- [ ] Practice your build steps one more time with paper.
- [ ] Memorize the mass of 1 nut (about 3 g) and your real-world percent.
- [ ] Redo any practice test question you missed.
- [ ] Get a good night's sleep.

Rules to remember

- Only kit materials in the barge.
- The cargo area must be open at the top.
- No scooping out water during testing.
- Loading stops at 500 g.

At the event

- [ ] Build first: 15 minutes, kit only, fold the corners, tape the seams.
- [ ] Box check: fits in 15 x 15 x 2.5 cm, open top.
- [ ] Write your estimate to 0.01 g BEFORE testing.
- [ ] Work on the written test until the judge calls you.
- [ ] Goggles on. Watch the judge load, then go back and finish the test.

After: my tournament report

Our score or time:

What went great:

What we would change:

Thank you to:

Draw or write my favorite moment of the day

Build Missions

One mission for each Ship Shape meeting. When you finish, show a coach or your Team Captain for initials, then color in the badge on page 19 and in your Team Badge Passport.

When	Mission	Badge	Done	Initials
After Fri Oct 23 Lesson 1	Penny Barge: load a foil boat, then beat it with a new shape (p. 13).	Penny Loader	☐	
After Fri Dec 4 Lesson 2	Capacity Predictor: predict, then test, with 1 cubic cm = 1 g (p. 14).	Archimedes Ace	☐	
After Fri Jan 22 Lesson 3	15-Minute Home Kit Build that fits the box (p. 15).	15-Minute Builder	☐	
After Fri Feb 19 Lesson 4	Estimate to the 0.01 g and score yourself (p. 16).	Estimate Expert	☐	
Anytime Plan	Sketch Design A and Design B on graph paper (p. 5-6).	Design Duelist	☐	
Anytime Log	Log 12 trials (p. 7-10) and graph your best cargo (p. 11).	Trial Tracker, Graph Maker	☐	
Anytime Bonus	Fix 3 problems (p. 12). Goggles on for every test and bag packed (p. 17).	Leak Stopper, Safety Captain	☐	

Tournament day: Sat Mar 13, Chatham Regional (Seaforth High School)

- ☐ Category C goggles
- ☐ Scissors, ruler, pencils
- ☐ Calculator (optional)

- Full checklist on page 17.
- Estimate first, to 0.01 g.
- Fold corners, spread the cargo.

My Badges

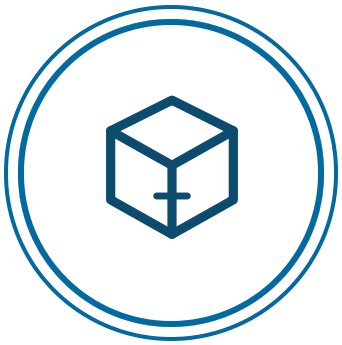
Color in each badge when you earn it. Earn all 9 to become a **Master Barge Builder**! Your Team Badge Passport has the same badges.



Penny Loader

Foil boat: load pennies, try 2 shapes

Initials: _____



Archimedes Ace

Predict capacity with 1 cubic cm = 1 g

Initials: _____



15-Minute Builder

Home-kit barge in 15 min that fits the box

Initials: _____



Estimate Expert

Estimate to 0.01 g, then test it

Initials: _____



Design Duelist

Sketch Design A and Design B

Initials: _____



Trial Tracker

Log 12 trials in the Trial Log

Initials: _____



Graph Maker

Graph your best cargo held

Initials: _____



Leak Stopper

Fix 3 problems on What went wrong?

Initials: _____



Safety Captain

Goggles on every test; pack your bag

Initials: _____